Total No. of Questions : 4]

SEAT No. :

PE-543

[Total No. of Pages : 2

[6578]-16

S.E. (Electronics & Computer Engineering) (Insem.)

ELECTRONIC CIRCUITS

(2019 Pattern) (Semester - III) (204202)

Time : 1 Hour] *[Max. Marks : 30*

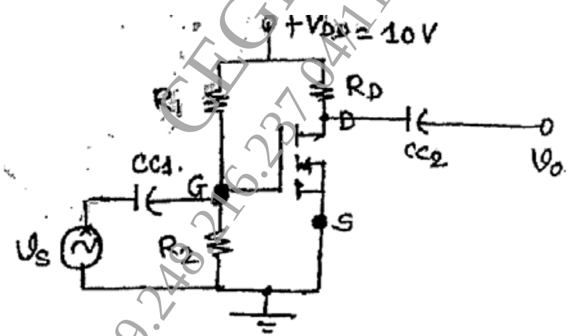
Instructions to the candidates :

1) ***Answer Q.1 or Q.2, & Q.3 or Q.4***
2) ***Figures to the right side indicates full mark.***
3) ***Draw neat diagram wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) Explain drain & transfer characteristics of N- EMOSFET. **[5]**

b) Explain any two non-ideal current voltage characteristics of MOSFET transistor. **[5]**

c) For the circuit diagram shown in figure 1, calculate V_{DS}, I_D & V_{GS}. **[5]**



Assume : R1 = 10 MΩ, R2 = 3.6 MΩ, Kn = 0.5 mA/V2, VTN = 1.5 V

OR

Q2) a) Draw common source of E-MOSFET amplifier and explain its modes of operation. **[5]**

b) Draw the ac equivalent circuit of a CS MOSFET amplifier and draw its small signal equivalent. **[5]**

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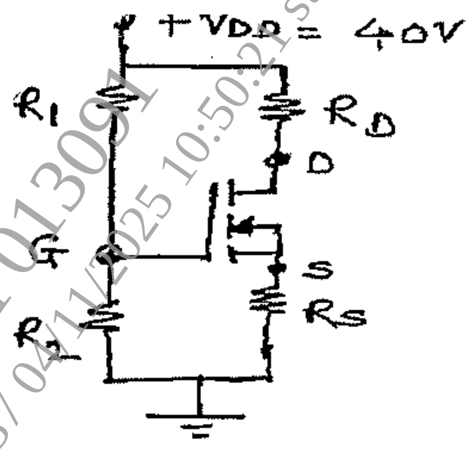
c) For the circuit shown in figure 2. Calculate I_{DQ}, V_{DSQ} and V_D. Assume : R_1 = 22 MΩ, R_D = 3 kΩ, R_2 = 18 MΩ, V_{TN} = 3 V, R_S = 0.82 kΩ Kn = 0.12 mA/V2, V_{GS} = 10.48 V **[5]**



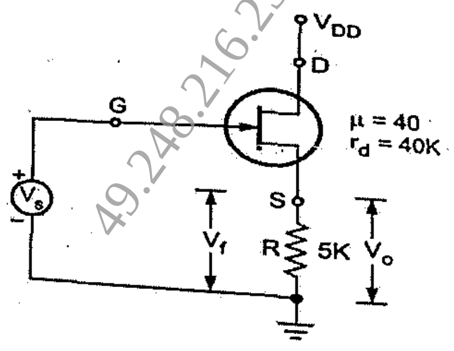
Q3) a) Mention the effect of negative feedback on amplifier's performance such as 1. Gain 2.i/p and o/p impedance 3. Sensitivity 4. Band width. 5. Stability. **[5]**

b) Draw circuit diagram of wein bridge oscillator and explain its working. **[5]**

c) Explain MOSFET as Diode/Resistor **[5]**

OR

Q4) a) Identify topology of feedback and determine A_{vf}, Rif, Rof for the amplifier shown in figure. For the MOSFET μ = 40, rd = 40 K. **[5]**



b) Draw circuit diagram of Phase shift Oscillator and explain its working. **[5]**

c) Compare different types of feedback topologies with respect to different parameter. **[5]**

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